

Efficient quantile regression under censoring using Laguerre polynomials

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In this paper, we consider a novel methodology to estimate linear quantile regression models when the response is randomly right censored. The proposed methodology is based on approximating the error distribution by means of an extension of the Laplace distribution, that is flexible enough to approximate any continuous distribution as long as the number of parameters in this Enriched Laplace distribution grows to infinity. The extension is obtained by enriching the Laplace density by means of Laguerre polynomials in such a way that the new density has by construction the property that its quantile of interest is equal to zero. Hence, when the error term has an Enriched Laplace density, it satisfies by construction the constraints of a quantile regression model. We will show that, with this new quantile regression model, we can obtain novel estimators of the quantile function, which are shown to be consistent and asymptotically normal. We also establish the asymptotic efficiency bound, and show by means of a simulation study and the analysis of data on Covid-19 patients that the proposed method works well in practice compared to competing estimators.

Keywords: Censored data; Laguerre polynomials; quantile regression; semi-parametric regression

1. Introduction

Consider the following linear quantile regression model:

$$T = \beta'_\tau X + \varepsilon, \quad (1.1)$$

where T is a response variable, X is a vector of covariates, β_τ is a vector of regression coefficients, and ε is an error term satisfying $\mathbb{P}(\varepsilon \leq 0|X) = \tau$ for some $0 < \tau < 1$. Quantile regression is becoming increasingly popular when one is not only interested in analyzing the center of a distribution (conditional on covariates) but also in analyzing the tail behavior since the tails might reveal important information about the effect of a covariate on a response. See [Koenker \(2017\)](#) for a recent review of quantile regression methodology. In the context of survival analysis, where the response is typically subject to right censoring, quantile regression is very much used and is often preferred over mean regression. This is not only because of the above advantages of quantile regression in general, but also because the nonparametric estimation of the mean is often not possible when the data are right censored. Moreover, survival data are often skewed to the right, making quantile regression even more appropriate, since they are not sensitive to outliers contrary to the mean.

The seminal paper by [Koenker and Bassett \(1978\)](#) shows that

$$\beta_\tau = \operatorname{argmin}_\beta \mathbb{E}[\rho_\tau(T - \beta'X)], \quad (1.2)$$

where $\rho_\tau(u) = u(\tau - \mathbb{1}(u \leq 0))$ is the so-called check loss function. This characterization of the quantile regression function is attractive since it allows to express the vector β_τ via the minimization of an expected loss function. Moreover, the minimization in (1.2) is equivalent to the maximization of the

following expected log-likelihood:

$$\beta_\tau = \operatorname{argmax}_\beta \mathbb{E}[\log f(T - \beta' X)], \quad (1.3)$$

where $f(t) = \tau(1 - \tau) \exp[-\rho_\tau(t)]$ is the asymmetric Laplace density. Hence, quantile regression can also be expressed via the maximization of an expected log-likelihood.

However, quantile regression coefficients cannot be expressed by minimizing an expected check loss or maximizing an expected log asymmetric Laplace likelihood when the response T or the covariates X are incomplete or observed with error. This is, e.g., the case when T or (some components of) X are subject to censoring of any kind (right, left, interval, double, twice,...), or to measurement errors (e.g. classical, Berkson,...). In all these cases, the vector β_τ in (1.2) or (1.3) does not characterize the quantile of T given X . We refer, e.g., to [Koenker \(2011\)](#), who showed this in a comment to the paper by [Bottai and Zhang \(2010\)](#) for the special case where T is randomly right censored.

Hence, in the literature on quantile regression with right censored responses, numerous methods have been proposed to correct for the censoring in T . The earlier contributions, like e.g. [Powell \(1984\)](#), [Powell \(1986\)](#), supposed that the censoring time is known for all subjects under study, so also the uncensored ones. This can be a restrictive assumption in many medical applications, where censoring does not only happen at the end of the study. Later, [Ying, Jung and Wei \(1995\)](#), [Bang and Tsiatis \(2002\)](#), and [Zhou \(2006\)](#) considered the more common case where the censoring time C is only observed for censored subjects, but they assumed that C is independent of the survival time T and the vector of covariates X . The latter assumption has subsequently been relaxed to conditional independence between T and C given X by a number of authors. The group of papers that worked under the latter assumption can basically be divided in two groups: some papers (like [Portnoy \(2003\)](#), [Peng and Huang \(2008\)](#) and [Yang, Narisetty and He \(2018\)](#)) assumed a globally linear quantile regression model (in the sense that the model is not only linear for the quantile level of interest, but also for all smaller quantile levels), whereas some other papers (like [Wang and Wang \(2009\)](#), [Leng and Tong \(2014\)](#), and [De Backer, El Ghouch and Van Keilegom \(2019, 2020\)](#)) only assumed linearity of the quantile regression function for the quantile level of interest. Finally, [Lee, Sit and Xu \(2023\)](#) proposed a unified framework that covers several of the aforementioned approaches as special cases. They also construct an efficient estimator of the regression parameters among all estimators that fall under this framework. A comprehensive review about quantile regression in survival analysis can be found in [Peng \(2021\)](#).

Contrary to the existing papers in the literature, we tackle the presence of imperfect measurements of the response by replacing the asymmetric Laplace density in (1.3) by an extension of the Laplace density, that is flexible enough to approximate any continuous distribution as long as the number of parameters in this Enriched Laplace distribution grows to infinity. The extension is obtained by enriching the Laplace density by means of Laguerre polynomials in such a way that the new density has by construction the property that its quantile of interest is equal to zero. Hence, when the error term has an Enriched Laplace density, it satisfies by construction the constraints of a quantile regression model. We exemplify our methodology in the situation of right censoring; we will show that, with this new quantile regression model, we can obtain novel estimators of the quantile function, which are shown to be consistent and asymptotically normal. We also establish the asymptotic efficiency bound for quantile regression in censored data, which has, to the best of our knowledge, never been obtained before, despite the abundance of papers on quantile regression with censored data.

While we focus in this paper on right-censoring, an important strength of the methodology that we will present is that it can be readily extended to other scenarios, in which, e.g., the response has other types of censoring patterns, the covariates are censored, there are measurement errors, missing data, etc. So in that sense, the proposed method will be versatile and can be extended to many other contexts. This is because our approach is based on the likelihood, which can easily handle different types of

data structures. This is in contrast to many existing quantile regression methods, that are often based on estimating equations derived from the check function or on other non-likelihood based estimating equations, and that can therefore not easily handle other types of incomplete data.

The paper is organized as follows. In the next section, we introduce the model, the proposed estimator, and a number of notations and assumptions. Section 3 is devoted to the development of the asymptotic efficiency bound, whereas the asymptotic theory for the proposed estimators is obtained in Section 4; this includes the consistency, rate of convergence, asymptotic normality and efficiency of the estimators. In Section 5, we discuss the implementation of the procedure, including the computation of bootstrap confidence intervals (Section 5.1), we provide an extensive simulation study (Section 5.2), and we show how our method can be applied to a real-world data-set in Section S.5.2 of the Supplementary Material Kreiss and Van Keilegom (2025). Finally, we conclude with some ideas for future research in Section 6. Most of the proofs are deferred to the Supplementary Material.

2. Notation, setting and estimator

Let T be a real-valued survival time and let X be a vector of random covariates taking values in $X \subseteq \mathbb{R}^{1+p}$. Our interest lies in the estimation of the τ -th conditional quantile function $m_\tau(X)$ of T given X , which we suppose to be linear, i.e.,

$$\mathbb{P}(T \leq m_\tau(X)|X) = \mathbb{P}(T \leq \beta'_\tau X|X) = \tau$$

for some $\beta_\tau \in \mathbb{R}^{1+p}$. For notational convenience, we will assume throughout that the first entry of $X = (1, X^{(1)}, \dots, X^{(p)})$ is an intercept. In addition, we consider right-censored data, that is, T is not observed directly, but instead we observe a censored survival time $Y := \min(T, C)$, where C is the (unknown) censoring time. We also observe $\Delta := \mathbb{1}(T \leq C)$, which indicates if Y is the actual survival time ($\Delta = 1$) or if it is a censored observation ($\Delta = 0$). Throughout, we will assume the following:

Assumption (A1). For $n \in \mathbb{N}$, $(T_i, C_i, X_i)_{i=1, \dots, n}$ are iid copies of (T, C, X) as above. Moreover, T and C are real-valued random variables, that are independent conditionally on X (taking values in $X \subseteq \mathbb{R}^{1+p}$), and there is $\beta_\tau \in \mathbb{R}^{1+p}$ such that $m_\tau(x) = \beta'_\tau x$. Furthermore, $\mathbb{E}(\|X\|^2) < \infty$, the conditional distributions of T given X and of C given X have densities $f_{T|X}$ and $f_{C|X}$, respectively (both with respect to the Lebesgue measure λ), and there is a measure μ on X such that the distribution of X has a density f_X with respect to a measure μ .

Note that this set-up explicitly allows for discrete and continuous (or both) covariates, depending on the choice of μ . Also note that we only assume linearity of $m_\tau(x)$ for the quantile level τ of interest. The following lemma is a direct consequence of (A1) and we omit the simple proof.

Lemma 2.1. Under (A1), the joint density of (T, X, C) with respect to $\lambda^2 \otimes \mu$ is given by

$$(t, x, c) \mapsto f_{T|X}(t|x) f_{C|X}(c|x) f_X(x). \quad (2.1)$$

We formulate now precisely which data we assume to observe:

Observations. Let Y_i and Δ_i be defined analogously to Y and Δ above. We suppose to observe $(Y_i, \Delta_i, X_i)_{i=1, \dots, n}$.

Under Assumption (A1), the factorized joint density of (T, X, C) given in (2.1) provides an expression for the joint density of (Y, X, Δ) with respect to $\lambda \otimes \mu \otimes \zeta$, where ζ denotes the counting measure:

$$h_0(y, x, \delta) := f_X(x) (\delta f_{T|X}(y|x)(1 - F_{C|X}(y|x)) + (1 - \delta) f_{C|X}(y|x)(1 - F_{T|X}(y|x))), \quad (2.2)$$

where capital letters refer to the corresponding distribution functions. The parameter of interest, namely β_τ , appears only in $f_{T|X}$ and $F_{T|X}$. Hence, we try to find a conditional density g of T given X which fits the data best. We do this by maximum likelihood estimation, where the likelihood of the observations $(Y_i, \Delta_i, X_i)_{i=1, \dots, n}$ can be found by taking the log of the density given in (2.2) and replacing $f_{T|X}$ by g . Up to an additive constant it is given by (let G denote the distribution function of g)

$$\mathcal{L}_n(g) := \sum_{i=1}^n \Delta_i \log g(Y_i|X_i) + (1 - \Delta_i) \log (1 - G(Y_i|X_i)). \quad (2.3)$$

Since maximization of $\mathcal{L}_n$ over the space of all densities is impractical, we study a sieve estimator: We will define finite dimensional function classes, the sieve spaces, comprising functions $g_{\beta, \gamma}$, where $\beta \in \mathbb{R}^{1+p}$ describes the linear conditional τ -th quantile function of $g_{\beta, \gamma}$ and $\gamma \in \mathbb{R}^{d_n}$ is a further parameter, which dimension d_n increases with n . Thus, with n growing, the function class becomes richer. We then consider estimators

$$(\hat{\beta}_n, \hat{\gamma}_n) = \underset{(\beta, \gamma)}{\operatorname{argmax}} \mathcal{L}_n(g_{\beta, \gamma}),$$

treating $\hat{\gamma}_n$ as nuisance parameter. This procedure works well if the true density can be well approximated by the sieve spaces.

Remark 2.2 (Extension to general forms of imperfect data). While we have discussed right-censoring above, we emphasize that this methodology can be directly extended to other forms of imperfect measurements. Suppose that one observes (W, X) , where W is an (almost) arbitrary function of T and possibly other (latent) random variables. In the case of right-censoring, we have $W = (T \wedge C, \mathbb{1}(T \leq C))$. The density of (W, X) takes the form $(w, x) \mapsto \Psi(f_{T|X}; w, x)$ for a suitable transformation Ψ of the density of interest $f_{T|X}$. When observing iid pairs $(W_i, X_i)_{i=1}^n$, we can hence proceed as described above, but using $\mathcal{L}_n(g) = \sum_{i=1}^n \log \Psi(g; W_i, X_i)$ instead of (2.3), as long as β can be interpreted as the quantile parameter. Therefore, we expect that other imperfect data scenarios of interest can be analyzed along the same lines as in this paper. We have, e.g., used such an idea in a completely different imperfect data context (in epidemics) in Kreiss and Van Keilegom (2022). We consider this conceptual flexibility a great strength of our method. The Kaplan Meier Estimator, e.g., is tailored to right-censored data, but it is unclear how to extend it to other scenarios.

We introduce now the sieve spaces and explain how to guarantee interpretability of β . To fix ideas, consider the special case $T = \beta'_\tau X + \sigma(X)\varepsilon$, where ε is independent of X and has its τ -th quantile at 0. Let f_ε be the density of ε with respect to λ . In this case, we can compute that $f_{T|X}(t|x)$ is given by

$$f_{T|X}(t|x) = \frac{1}{\sigma(x)} f_\varepsilon \left(\frac{t - \beta'_\tau x}{\sigma(x)} \right). \quad (2.4)$$

The idea of our estimator is to non-parametrically estimate the best approximation of a general $f_{T|X}$ in terms of (2.4). Thus, we can expect our estimator to work well if the true conditional density $f_{T|X}$ can

be approximated well by (2.4) for suitable choices of σ and f_ε . In Theorem 4.2 below, this is encoded in the assumption $\delta_n(\alpha) \rightarrow 0$. To define finite dimensional approximations to f_ε and σ , let

$$L_k : [0, \infty) \rightarrow \mathbb{R}, \quad x \mapsto \sum_{i=0}^k \binom{k}{i} \frac{(-x)^i}{i!}$$

be the Laguerre polynomial of order $k \in \mathbb{N}$ by L_k . Denote furthermore by $f_\rho(x) = \rho e^{-\rho x} \mathbb{1}(x \geq 0)$ the density of an exponential distribution with rate $\rho > 0$. We omit the simple proof of the following lemma, which is based on the following fact about Laguerre polynomials: $\int_0^\infty L_{k_1}(x) L_{k_2}(x) e^{-x} dx = \delta_{k_1, k_2}$, where $\delta(k_1, k_2) = 1$ if $k_1 = k_2$ and 0 else (below $\|v\|$ for a real vector v denotes its Euclidean norm).

Lemma 2.3. *For any $\tau \in (0, 1)$, $m, \tilde{m} \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$, $\theta \in \mathbb{R}^{m+1}$, and $\tilde{\theta} \in \mathbb{R}^{\tilde{m}+1}$, the function*

$$f_{\theta, \tilde{\theta}}(u) := (1 - \tau) f_\tau(u) \left(\sum_{k=0}^m \frac{\theta_k}{\|\theta\|} L_k(\tau u) \right)^2 + \tau f_{1-\tau}(-u) \left(\sum_{k=0}^{\tilde{m}} \frac{\tilde{\theta}_k}{\|\tilde{\theta}\|} L_k(-(1-\tau)u) \right)^2 \quad (2.5)$$

is the density of a probability distribution whose quantile of level τ lies at 0. Denote by $F_{\theta, \tilde{\theta}}$ the corresponding distribution function.

Similarly, we also consider polynomial approximations to σ . Let

$$H_k(x) := \frac{1}{\sqrt{k!}} (-1)^k e^{\frac{x^2}{2}} \frac{d^k}{dx^k} e^{-\frac{x^2}{2}}$$

denote the k -th Hermite polynomial. They have the property that $\int_{-\infty}^\infty H_{k_1}(x) H_{k_2}(x) \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \delta_{k_1, k_2}$. Therefore, we obtain the following analogous result to Lemma 2.3:

Lemma 2.4. *For all $M \in \mathbb{N}_0$ and all $\lambda \in \mathbb{R}^{M+1}$, the function*

$$\sigma_\lambda(x) := \left(\sum_{k=0}^M \frac{\lambda_k}{\|\lambda\|} H_k(x) \right)^2$$

satisfies $\int_{-\infty}^\infty \sigma_\lambda(x) \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = 1$.

Let $x = (1, x^{(1)}, \dots, x^{(p)})$ and $\lambda = (\lambda_1, \dots, \lambda_p)$, where $\lambda_a \in \mathbb{R}^{M+1}$ for $a = 1, \dots, p$. Let $\sigma_\lambda(x) := \prod_{a=1}^p \sigma_{\lambda_a}(x^{(a)})$. We have now all definitions to define the sieve spaces of conditional densities. We denote for $m, \tilde{m}, M \in \mathbb{N}_0$ and a compact set $K \subseteq \mathbb{R}^{1+p}$ the spaces of densities

$$\mathcal{F}_{m, \tilde{m}, M} := \left\{ g_{\theta, \tilde{\theta}, \lambda, \beta}(t|x) := \frac{1}{\sigma_\lambda(x)} f_{\theta, \tilde{\theta}} \left(\frac{t - \beta'x}{\sigma_\lambda(x)} \right) \mid \theta \in \mathbb{R}^{m+1}, \tilde{\theta} \in \mathbb{R}^{\tilde{m}+1}, \lambda \in \mathbb{R}^{p \times (M+1)}, \beta \in K \right\}.$$

We do not add the dependence on K in the notation because we will not be concerned with its choice, but assume the existence of a sequence of compact K_n such that $\beta_\tau \in K_n$ for $n \geq N$ and some $N \in \mathbb{N}$. Note furthermore that, in the case $p = 0$, the function $\sigma_\lambda(x)$ would be a constant; thus, we do not need to include λ , the model is only indexed via θ and $\tilde{\theta}$, and we do not have to choose M .

For given sequences $(m_n), (\tilde{m}_n), (M_n) \subseteq \mathbb{N}_0$ and compact sets $K_n \subseteq \mathbb{R}^{1+p}$, we obtain the sieve maximum likelihood estimator by maximizing $\mathcal{L}_n$ over $\mathcal{F}_{m_n, \tilde{m}_n, M_n}$ or equivalently by maximizing

$$L(\beta, \theta, \tilde{\theta}, \lambda) := \sum_{i=1}^n \Delta_i \log g_{\theta, \tilde{\theta}, \lambda, \beta}(Y_i | X_i) + (1 - \Delta_i) \log \left(1 - G_{\theta, \tilde{\theta}, \lambda, \beta}(Y_i | X_i) \right) \quad (2.6)$$

over $\beta, \theta, \tilde{\theta}, \lambda$, i.e., the estimator is given by

$$(\hat{\beta}_n, \hat{\theta}_n, \hat{\tilde{\theta}}_n, \hat{\lambda}_n) := \underset{\substack{\beta \in K_n, \lambda \in \mathbb{R}^{p \times (M_n+1)} \\ \theta \in \mathbb{R}^{m_n+1}, \tilde{\theta} \in \mathbb{R}^{\tilde{m}_n+1}}}{\operatorname{argmax}} L(\beta, \theta, \tilde{\theta}, \lambda).$$

Our interest lies in $\hat{\beta}_n$, and $(\hat{\theta}_n, \hat{\tilde{\theta}}_n, \hat{\lambda}_n)$ will be treated as estimators of nuisance parameters. The tuning parameters $m_n, \tilde{m}_n$ and M_n can be found via cross-validation (cf. Section 5.1 for more details). Note that ε and σ in (2.4) are not uniquely determined. While this is in principle not problematic because in all cases the conditional quantile (our interest) remains the same, it will be in the interest of numerical stability to take care of this problem. We come back to the issue in the implementation Section 5.1.

Remark 2.5.

1. One reason, why σ and f_ε in (2.4) are not unique, is because $\tilde{\sigma}(\cdot) := \gamma^{-1} \sigma(\cdot)$ and $\tilde{f}_\varepsilon(\cdot) := \gamma^{-1} f_\varepsilon(\cdot/\gamma)$ yield the same joint density for all $\gamma > 0$. Lemma 2.4 shows that our parametrisation of σ naturally imposes a constraint on the scaling and therefore avoids this issue.
2. For any $\gamma \neq 0$, θ and $\gamma\theta$ (and likewise $\tilde{\theta}$ and $\gamma\tilde{\theta}$ as well as λ and $\gamma\lambda$) yield the same densities (or σ). Theoretically, this will be not important since the quantile is unaffected. But we use spherical coordinates in Section 5.1 to avoid any issues caused by this ambiguity in the implementation.

Note finally that

$$\hat{g}_n(t|x) := g_{\hat{\theta}_n, \hat{\tilde{\theta}}_n, \hat{\lambda}_n, \hat{\beta}_n}(t|x) = \frac{1}{\sigma_{\hat{\lambda}_n}(x)} \hat{f}_n\left(\frac{t - \hat{\beta}'_n x}{\sigma_{\hat{\lambda}_n}(x)}\right), \text{ where } \hat{f}_n(u) := f_{\hat{\theta}_n, \hat{\tilde{\theta}}_n}(u),$$

can be seen as an estimator for $f_{T|X}$. Thus, we can consider other properties of $f_{T|X}$ by estimating them by the corresponding properties of $\hat{g}_n(t|x)$. Let, e.g., $\hat{q}_n(\tau')$ be the τ' -th quantile of $\hat{f}_n$. The τ' -th conditional quantile of $\hat{g}_n(t|x)$ is hence given by $x' \hat{\beta}_n + \sigma_{\hat{\lambda}_n}(x) \hat{q}_n(\tau')$. Note that only for $\tau' = \tau$ (in which case $\hat{q}_n(\tau) = 0$) the estimated quantile function is linear unless $\sigma_{\hat{\lambda}_n}(x)$ is linear. Thus, in principle, one estimator can be used to estimate several quantiles simultaneously avoiding quantile crossing. However, we will not provide theory for this estimator. In our theory, $\beta, \theta, \tilde{\theta}, \lambda$ depend on τ and have to be re-estimated if τ changes.

In the following, $\rho(h_1, h_2)$ denotes the Hellinger distance between two densities h_1, h_2 on $\mathbb{R} \times \mathcal{X} \times \{0, 1\}$ with respect to $\lambda \otimes \mu \otimes \zeta$:

$$\rho(h_1, h_2) := \left(\int_{\mathbb{R} \times \{0, 1\} \times \mathcal{X}} \left(h_1(y, x, \delta)^{\frac{1}{2}} - h_2(y, x, \delta)^{\frac{1}{2}} \right)^2 dy d\zeta(\delta) d\mu(x) \right)^{\frac{1}{2}}.$$

3. Asymptotic efficiency bound

In this Section, we discuss the optimal asymptotic variance, i.e., the semi-parametric efficiency bound, for quantile estimators under censoring that use (A1) as only structural assumption. For this, we will

follow the framework of Chapter 25 in [van der Vaart \(1998\)](#). Proofs of the results in this section are provided in S.3 of the Supplementary Material ([Kreiss and Van Keilegom, 2025](#)).

We study quantile estimation without censoring first. To use the results later, we pretend to observe (T, C, X) (by conditional independence, for estimation of quantiles of T , this is the same as observing only T). We index the model by $f_{T|X}$, $f_{C|X}$, and f_X . Like in Example 25.16 in [van der Vaart \(1998\)](#), let $k(r) := 2(1 + e^{-2r})^{-1}$ and consider for given $(f_{T|X}, f_{C|X}, f_X)$ and $\delta > 0$ the following perturbations:

$$f_{\delta, T|X}(t) := c_{g_{T|X}}(\delta, x) k(\delta g_{T|X}(t|x)) f_{T|X}(t|x), \quad (3.1)$$

$$f_{\delta, C|X}(c) := c_{g_{C|X}}(\delta, x) k(\delta g_{C|X}(c|x)) f_{C|X}(c|x), \quad (3.2)$$

$$f_{\delta, X}(x) := c_{g_X}(\delta) k(\delta g_X(x)) f_X(x), \quad (3.3)$$

where $g_{T|X}$, $g_{C|X}$, and g_X are (for now) arbitrary functions. $k(\delta g_{T|X}(t|x))$ is bounded for any choice of $g_{T|X}$ (and similarly for $g_{C|X}$ and g_X). Hence, for suitable normalizing constants $c_{g_{T|X}}(\delta|x)$, $c_{g_{C|X}}(\delta, x)$, and $c_{g_X}(\delta)$, the functions defined in (3.1)-(3.3) are (conditional) probability density functions. The proof of the next Lemma is mostly about the behavior of the normalizing constants.

Lemma 3.1. *Let $g_{T|X}, g_{C|X} : \mathbb{R} \times \mathcal{X} \rightarrow \mathbb{R}$ and $g_X : \mathcal{X} \rightarrow \mathbb{R}$ be functions satisfying*

$$\begin{aligned} \int_{\mathcal{X}} g_X(x) f_X(x) d\mu(x) &= \int_{-\infty}^{\infty} g_{T|X}(t|x) f_{T|X}(t) dt = \int_{-\infty}^{\infty} g_{C|X}(c|x) f_{C|X}(c|x) dc = 0 \text{ for all } x \in \mathcal{X}, \\ \int_{\mathcal{X}} \left(g_X(x)^2 + \int_{-\infty}^{\infty} g_{T|X}(t|x)^2 f_{T|X}(t|x) dt + \int_{-\infty}^{\infty} g_{C|X}(c|x)^2 f_{C|X}(c|x) dc \right) f_X(x) d\mu(x) &< \infty. \end{aligned} \quad (3.4)$$

The derivative in quadratic mean, i.e., the score function, of the sub-model $\delta \mapsto (f_{\delta, T|X}, f_{\delta, C|X}, f_{\delta, X})$ defined in (3.1)-(3.3) is given by $(t, c, x) \mapsto g_{T|X}(t|x) + g_{C|X}(c|x) + g_X(x)$. Moreover, the score function of an arbitrary sub-model $\delta \mapsto (f_{\delta, T|X}, f_{\delta, C|X}, f_{\delta, X})$ is of this type if: $\delta \mapsto \sqrt{f_{\delta, T|X}}$, $\delta \mapsto \sqrt{f_{\delta, C|X}}$, and $\delta \mapsto \sqrt{f_{\delta, X}}$ are continuously differentiable, and the following are continuous: $\delta \mapsto \mathbb{E}_{\delta}((\frac{d}{d\delta} \log f_{\delta, T|X}(t|x) f_{\delta, C|X} f_{\delta, X}(x))^2)$, $\delta \mapsto \int_{-\infty}^{\infty} (d/d\delta \log f_{\delta, T|X}(t,x))^2 f_{\delta, T|X}(t|x) dt$ and similarly for $f_{\delta, C|X}$ and $f_{\delta, X}$. When talking about tangent spaces below, we refer to such sub-models.

Having this result, we can find the efficient influence function and the semi-parametric efficiency bound for the case of uncensored observations.

Lemma 3.2. *Let (A1) hold. Suppose that $X'\gamma = 0$ a.s. $\Rightarrow \gamma = 0$ and that the following integral is well-defined: $\int_{\mathcal{X}} f_X(x) f_{T|X}(x'\beta_{\tau}|x)^{-2} x x' d\mu(x)$. Then, the matrix $\Sigma := \int_{\mathcal{X}} x x' f_X(x) d\mu(x)$ is invertible, and the efficient influence function for estimation of β_{τ} from complete observations $(T_i, X_i), i = 1, \dots, n$ is given by*

$$\Psi(t, x) := \frac{\tau - \mathbb{1}(t \leq x'\beta_{\tau})}{f_{T|X}(x'\beta_{\tau}|x)} \Sigma^{-1} x.$$

In particular, this implies that all regular estimators have a variance no lower than

$$\mathbb{E}(\Psi(T, X) \Psi(T, X)') = \tau(1 - \tau) \Sigma^{-1} \int_{\mathcal{X}} \frac{f_X(x)}{f_{T|X}(x'\beta_{\tau}|x)^2} x x' d\mu(x) \Sigma^{-1}.$$

After this preliminary discussion, we turn to the case of censored observations (Y, Δ, X) defined as before. As step one, we compute the score functions in the model with censoring.

Lemma 3.3. Let $\delta \mapsto (f_{\delta,T|X}, f_{\delta,C|X}, f_{\delta,X})$ be defined as in (3.1)-(3.3), where $g_{T|X}, g_{C|X} : \mathbb{R} \times \mathcal{X} \rightarrow \mathbb{R}$ and $g_X : \mathcal{X} \rightarrow \mathbb{R}$ fulfill (3.4). The score for observing (Y, Δ, X) is given by $(y \in \mathbb{R}, x \in \mathcal{X} \text{ and } b \in \{0, 1\})$

$$(y, b, x) \mapsto b \left(g_{T|X}(y|x) + g_X(x) + \frac{\int_y^\infty g_{C|X}(c|x) f_{C|X}(c|x) dc}{1 - F_{C|X}(y|x)} \right) \\ + (1 - b) \left(g_{C|X}(y|x) + g_X(x) + \frac{\int_y^\infty g_{T|X}(t|x) f_{T|X}(t|x) dt}{1 - F_{T|X}(y|x)} \right).$$

In particular, the tangent space at $(f_{T|X}, f_{C|X}, f_X)$ with respect to the sub-models defined in (3.1)-(3.3) is given as the collection of all functions of the above form, where $g_{T|X}, g_{C|X}, g_X$ range over all functions fulfilling (3.4).

We can now formulate the main result of this section.

Theorem 3.4. Let (A1) hold and suppose that $f_{T|X}(X'\beta_\tau|X) > 0$ and $F_{C|X}(X'\beta_\tau|X) < 1$ almost surely. Define $c_0(x) := -(1 - \tau)/f_{T|X}(x'\beta_\tau|x)$,

$$I(u, x) := \int_{-\infty}^u \frac{f_{C|X}(r|x)}{(1 - F_{T|X}(r|x))(1 - F_{C|X}(r|x))^2} dr,$$

$$J(u, x) := - \int_{-\infty}^u \frac{f_{T|X}(r|x)}{(1 - F_{T|X}(r|x))^2(1 - F_{C|X}(r|x))} dr,$$

$$\Psi_1(y, x) := c_0(x) \Sigma^{-1} x (\mathbb{1}(y \leq x'\beta_\tau) (I(y, x) + 1) + \mathbb{1}(y > x'\beta_\tau) J(x'\beta_\tau, x)), \text{ and}$$

$$\Psi_0(y, x) := c_0(x) \Sigma^{-1} x (\mathbb{1}(y \leq x'\beta_\tau) J(y, x) + \mathbb{1}(y > x'\beta_\tau) J(x'\beta_\tau, x)).$$

If $\mathbb{E}(\Psi_1^{(k)}(Y, X)^2) < \infty$ for all $k = 1, \dots, p + 1$, then the efficient influence function for estimating the τ -th quantile from censored observations is given by $\Psi : \mathbb{R} \times \{0, 1\} \times \mathcal{X} \rightarrow \mathbb{R}$, $(y, b, x) \mapsto \Psi_b(y, x) = b\Psi_1(y, x) + (1 - b)\Psi_0(y, x)$.

Remark 3.5. The assumption $F_C(X'\beta_\tau|X) < 1$ a.s. means that with positive probability some censoring times lie behind the quantile. It is clear that otherwise no inference is possible. Note moreover that in the definitions of Ψ_1 and Ψ_0 the integrals $I(y, x)$ and $J(y, x)$ are only considered for $y \leq x'\beta_\tau$. This implies that $1 - F_{T|X}(y|x) \geq 1 - \tau$. Hence, both integrals are well defined on this interval.

We summarize all results of this section in the following corollary of Theorem 3.4. Denote the τ -th conditional quantile of $f_{\delta,T|X}$ defined as in (3.1) by $\beta_{\delta,\tau}(g_{T|X}, g_{C|X}, g_X)$ (cf. Section 4.2 in (4.6) for how to define it if $f_{\delta,T|X}$ has no quantile function linear in X). An estimator T_n of β_τ is called regular if there is a distribution L such that for any submodel in (3.1)-(3.3) $\sqrt{n}(T_n - \beta_{1/\sqrt{n},\tau}(g_{T|X}, g_{C|X}, g_X)) \rightarrow L$ in distribution, assuming that $(Y_i, X_i, \Delta_i)_{i=1}^n$ has joint density $f_{1/\sqrt{n},T|X}(t|x)f_{1/\sqrt{n},C|X}(c|x)f_{1/\sqrt{n},X}(x)$.

Corollary 3.6. Let all conditions of Theorem 3.4 hold. The problem of estimating β_τ from censored observations is a semi-parametric problem. For any regular estimator T_n of β_τ with limiting distribution L , there exists a distribution M such that $L = N(0, v_0^2) * M$, where

$$v_0^2 := \mathbb{E}(\Psi_\Delta(Y, X) \Psi_\Delta(Y, X)')$$

$$= \Sigma^{-1} \mathbb{E} \left(\frac{(1-\tau)^2}{f_{T|X}(X'\beta_\tau|X)^2} XX' \left(\frac{\Delta \mathbb{1}(Y \leq X'\beta_\tau)}{(1-F_{T|X}(Y|X))(1-F_{C|X}(Y|X))} + J(Y \wedge X'\beta_\tau, X) \right)^2 \right) \Sigma^{-1}.$$

Moreover, for estimators T_n^* which assume knowledge of the distribution $F_{C|X}$ and F_X the same holds true. Therefore, it is the case that the censoring distribution is a true nuisance parameter in the sense that knowledge of the censoring distribution is not decreasing the optimal variance.

We consider two special cases (cf. Lemma S.3.3 in the Supplementary Material (Kreiss and Van Keilegom, 2025) for details).

Remark 3.7.

1. *Uncensored linear model.* Suppose that $T = x'\beta_\tau + \varepsilon$, i.e., $f_{T|X}(t|x) = f(t - x'\beta_\tau|x)$, and that $C = \infty$. According to Corollary 3.6, the optimal covariance matrix equals

$$\Sigma^{-1} \mathbb{E} \left(\frac{\tau(1-\tau)}{f(0|X)^2} XX' \right) \Sigma^{-1}.$$

Newey and Powell (1990) studies median ($\tau = 1/2$) regression in this model under a different censoring regime (known censoring). We can therefore compare with our results in the case of no censoring. Newey and Powell (1990) obtains $1/4 \cdot \mathbb{E}(f(0|X)^2 XX')^{-1}$ as semi-parametric efficiency bound. This is analogous to the bound in linear mean regression, $\mathbb{E}(XX'/f(0|X))^{-1}$, cf. Example 25.28 in van der Vaart (1998). For median regression, we obtain $\Sigma^{-1} \mathbb{E}(XX'/f(0|X)^2) \cdot \Sigma^{-1}/4$. The difference is due to the fact that we investigated the efficiency bound for estimators that make no use of a linear structure. Under homoskedasticity ($f(\varepsilon|x) = f(\varepsilon)$), however, we see that both variances coincide. This shows that, when estimating the median in a homoskedastic linear model without censoring, knowledge of this fact is not beneficial.

2. *No covariates.* If $X = 1$, the formula from Corollary 3.6 reduces to

$$\frac{(1-\tau)^2}{f_T(\beta_\tau)^2} \mathbb{E} \left(\left(\frac{\Delta \mathbb{1}(Y \leq \beta_\tau)}{(1-F_T(Y))(1-F_C(Y))} + J(Y \wedge \beta_\tau) \right)^2 \right).$$

Theorem 2 in Lo and Singh (1986) provides an asymptotic expansion of the inverse of the Kaplan Meier Estimator in the case of no covariates. One can show that the variance of this quantile estimator equals the above. Hence, inverting Kaplan-Meier is efficient for estimating quantiles under censoring. This is no surprise as Wellner (1982) shows that Kaplan-Meier is optimal.

4. Asymptotic theory

We analyze the asymptotic performance of our estimator for the case $M = M_n = 0$. Therefore, we will write in this chapter $\mathcal{F}_{m,\tilde{m}} := \mathcal{F}_{m,\tilde{m},0}$. While we do not suppose homoskedasticity in the data, the reader should note that below, e.g., in Theorem 4.2 or Assumption (AN2), we implicitly require that the true model can be reasonably well approximated by a homoskedastic model.

4.1. Consistency and rate of convergence

In this section, we study the rate of convergence in the Hellinger distance between the estimated density and the true density. From these, we will obtain a consistency result on the quantiles. The main

problem will be that the approximations $f_{\theta, \tilde{\theta}}$ can have zeros, which will result in singularities of the log-likelihood. Therefore, we cannot apply the standard results from empirical process theory. Instead, we will rely on [Wong and Shen \(1995\)](#). The proofs are given in Section S.1 of the Supplementary Material ([Kreiss and Van Keilegom, 2025](#)). Define

$$p_+(u|x) := \sqrt{\frac{f_{T|X}(u/\tau + x'\beta_\tau|x)}{\tau(1-\tau)}} e^u \text{ and } p_-(u|x) := \sqrt{\frac{f_{T|X}(-u/(1-\tau) + x'\beta_\tau|x)}{\tau(1-\tau)}} e^u.$$

For $y \neq x'\beta_\tau$ (since this happens with probability zero, we may ignore this condition in the following),

$$\begin{aligned} f_{T|X}(y|x) &= (1-\tau) \cdot f_\tau(y - x'\beta_\tau) p_+(\tau(y - x'\beta_\tau)|x)^2 \\ &\quad + \tau \cdot f_{1-\tau}(-(y - x'\beta_\tau)) p_-(-(1-\tau)(y - x'\beta_\tau)|x)^2. \end{aligned} \quad (4.1)$$

For given sequences $(m_n, \tilde{m}_n)$, let $g_n(y|x) = f_n(y - x'\beta) \in \mathcal{F}_{m_n, \tilde{m}_n}$ for some $f_n = f_{\theta_n, \tilde{\theta}_n}$. Then,

$$\begin{aligned} g_n(y|x) &= f_n(y - x'\beta) = (1-\tau) \cdot f_\tau(y - x'\beta) p_{+,n}(\tau(y - x'\beta))^2 \\ &\quad + \tau \cdot f_{1-\tau}(-(y - x'\beta)) p_{-,n}(-(1-\tau)(y - x'\beta))^2, \end{aligned} \quad (4.2)$$

where $p_{+,n}$ and $p_{-,n}$ are weighted sums of Laguerre polynomials. Note that $p_{+,n}$ and $p_{-,n}$ need not be considered conditional on x because in $\mathcal{F}_{m_n, \tilde{m}_n}$ the dependence on x is only through the shift $\beta'x$. This is analogous to the case where $f_{T|X}(y|x) = f_0(y - x'\beta_\tau)$. In this case, p_+ and p_- are independent of x .

From (4.1) and (4.2), we see that $f_n(y - x'\beta)$ will be a good approximation of $f_{T|X}$ provided that $p_{+,n}$ and $p_{-,n}$ are good approximations for p_+ and p_- , respectively. The densities f_n induce approximations to the true density of the observations h_0 as follows (F_n denotes the distribution function of f_n):

$$h(f_n, \beta|y, x, \delta) := [\delta f_n(y - x'\beta)(1 - F_{C|X}(y|x)) + (1 - \delta)(1 - F_n(y - x'\beta))f_{C|X}(y|x)] f_X(x).$$

For given sequences $(m_n), (\tilde{m}_n)$ and growing compact sets $K_n \subseteq \mathbb{R}^{p+1}$, we define, in addition to the spaces $\mathcal{F}_{m_n, \tilde{m}_n}$, also approximating spaces to the joint densities. More precisely, let

$$\mathcal{H}_n := \left\{ h(f_{\theta, \tilde{\theta}}, \beta|\cdot) : \beta \in K_n, \theta \in \mathbb{R}^{m_n+1}, \tilde{\theta} \in \mathbb{R}^{\tilde{m}_n+1} \right\}$$

be the space of all possible approximations of densities of the observed triples (Y, X, Δ) which are induced by the conditional densities in $\mathcal{F}_{m_n, \tilde{m}_n}$.

We need to discuss two main issues: Firstly, the size of the sieve spaces in terms of bracketing entropy (which we delay to Lemma S.1.1 in the Supplementary Material ([Kreiss and Van Keilegom, 2025](#))) and, secondly, the approximation quality of the sieve. Define to this end for any two densities p, q and for $\alpha \neq 0$,

$$\rho_\alpha(p, q) := \frac{1}{\alpha} \int_{\mathbb{R} \times \mathcal{X} \times \{0,1\}} p(y, x, \delta) \left[\left(\frac{p(y, x, \delta)}{q(y, x, \delta)} \right)^\alpha - 1 \right] dy d\mu(x) d\zeta(\delta).$$

Note that $\rho_{-1/2}$ equals the squared Hellinger-distance (see [Wong and Shen \(1995\)](#), page 351). We state the following general Lemma and discuss a special case afterwards.

Lemma 4.1. For any $g_n(y|x) = f_n(y - x'\beta) \in \mathcal{F}_{m_n, \tilde{m}_n}$, let $p_{+,n}$ and $p_{-,n}$ be defined as in (4.2), and let F_n denote the distribution function of f_n . For any $\alpha \in (0, 0.5)$, let

$$\begin{aligned} C_n(\alpha) &:= \sup_{x \in \mathcal{X}} \left(\int_{\mathbb{R}} f_{T|X}(y|x) \left(\frac{f_{T|X}(y|x)}{f_n(y - x'\beta_\tau)} \right)^{\frac{\alpha}{1-\alpha}} dy \right)^{1-\alpha}, \\ \tilde{C}_n(\alpha) &:= \sup_{x \in \mathcal{X}} \left(\int_{\mathbb{R}} \left(\frac{1 - F_{T|X}(y|x)}{1 - F_n(y - x'\beta_\tau)} \right)^{\frac{\alpha}{1-\alpha}} (1 - F_{T|X}(y|x)) f_{C|X}(y|x) dy \right)^{1-\alpha}, \\ \delta_{+,n}(x) &:= (1 - \tau) \int_0^\infty e^{-u} |p_+(u|x) - p_{+,n}(u)|^2 du, \text{ and} \\ \delta_{-,n}(x) &:= \tau \int_0^\infty e^{-u} |p_-(u|x) - p_{-,n}(u)|^2 du. \end{aligned}$$

Then, for $\alpha \in (0, 0.5)$,

$$\delta_n(\alpha) := \rho_\alpha(h_0, h(f_n, \beta_\tau|\cdot)) \leq \frac{1}{\alpha} \left(C_n(\alpha) + \tilde{C}_n(\alpha) \right) \int_{\mathcal{X}} (\delta_{+,n}(x)^\alpha + \delta_{-,n}(x)^\alpha) f_X(x) d\mu(x).$$

The main result of this section provides a convergence rate of the density estimate.

Theorem 4.2. Suppose (A1) holds, and let $(m_n), (\tilde{m}_n) \subseteq \mathbb{N}_0$ and $K_n \subseteq \mathbb{R}^{1+p}$ compact be given. Let $\delta_n(\alpha)$ be defined as in Lemma 4.1 with a sequence f_n such that there is an $\alpha \in (0, 0.5)$ with $\delta_n(\alpha) \rightarrow 0$. Let furthermore $\varphi_n \geq 0$ be a sequence such that (below $c > 0$ is the constant from Theorem S.4.1 in the Supplementary Material (Kreiss and Van Keilegom, 2025) and $|K_n|$ denotes the diameter of K_n)

$$\frac{\log |K_n|}{\varphi_n} + \frac{m_n + \tilde{m}_n}{\varphi_n} \log \left(\frac{n(m_n^3 + \tilde{m}_n^3)}{\varphi_n} \right) = O(1), \quad \text{and let} \quad r_n := \max \left(\sqrt{\frac{\varphi_n}{n}}, \sqrt{\frac{4\delta_n(\alpha)}{c}} \right).$$

Let $\hat{h}_n = h(f_{\hat{\theta}_n, \hat{\theta}_n}, \hat{\beta}_n|\cdot)$ be the estimated density of the observations. Then, $\mathbb{P}(\rho(h_0, \hat{h}_n) \geq r_n) \rightarrow 0$.

The following remarks discuss the assumption $\delta_n(\alpha) \rightarrow 0$ (which means that the true density can be well approximated by the Laguerre densities) and the rate of convergence.

Remark 4.3. Lemma 4.1 is useful if we can find a sequence $(f_n)_{n \in \mathbb{N}}$ for which $\delta_n(\alpha) \rightarrow 0$ for some α . Consider the special case $f_{T|X}(t|x) = \varphi_0(t - x'\beta_\tau)$; hence, $p_+(u|x) = \sqrt{\varphi_0(u/\tau)} e^u / \tau(1 - \tau)$ and $\delta_{+,n}(x)$ do not depend on x . Thus, we write simply $p_+(u)$ and $\delta_{+,n}$ instead. Similarly, we obtain expressions for p_- and $\delta_{-,n}$.

- We discuss the convergence behavior of $\delta_{+,n}$ under a mild condition on the right tail of φ_0 . The arguments below transfer in an analogous fashion to $\delta_{-,n}$ under a similar condition on the left tail of φ_0 . For this part of the remark only, we consider $\delta_{+,n}(p_{+,n})$ as function of the polynomial $p_{+,n}$. We will always suppose that $p_{+,n}$ is a weighted sum of Laguerre polynomials up to degree m_n . For the moment, we do not require that the weights fulfill any constraint. It is a known result (cf. Theorem 1 in Chapter II.8 of Nikiforov and Uvarov (1988)) that there is a sequence $\tilde{p}_{+,n}$ (a weighted sum of the first m_n Laguerre polynomials) such that $\delta_{+,n}(\tilde{p}_{+,n})$ converges to zero

whenever $m_n \rightarrow \infty$ and the following integrals are finite:

$$\int_0^\infty p_+(u)^2 e^{-u} du = \int_0^\infty \frac{\varphi_0(u/\tau)}{\tau(1-\tau)} du = 1 < \infty \quad \text{and}$$

$$\int_0^\infty (p'_+(u))^2 u e^{-u} du = \int_0^\infty \frac{\tau u}{4(1-\tau)} \left(\frac{\varphi'_0(u)^2}{\tau^2 \varphi_0(u)} + \varphi_0(u) + \frac{2\varphi'_0(u)}{\tau} \right) du < \infty.$$

While the first condition is always fulfilled, the second one poses a weak condition on the right tail of φ_0 . Let $\tilde{p}_{+,n}$ be the minimizer of $p \mapsto \delta_{+,n}(p)$, where p may range over all linear combinations of Laguerre polynomials up to degree m_n . In general, the weights of the Laguerre polynomials in $\tilde{p}_{+,n}$ will not sum to one when squared. By using Lagrange multipliers, it is possible to show that the minimizer of $\delta_{+,n}(p)$ among all p within our sieve-spaces (i.e., with the squares of the weights summing to one) is obtained by normalizing the coefficients of $\tilde{p}_{+,n}$, yielding a valid $p_{+,n}$. Moreover, it is clear that the sum of the coefficients of $\tilde{p}_{+,n}$ converges to one by Parseval's Inequality. Therefore, $\delta_{+,n} \rightarrow 0$.

- We discuss next the behavior of $C_n(\alpha)$ and $\tilde{C}_n(\alpha)$. We note first that under the above integral condition (and a version for the left tail) $p_{+,n}$ and $p_{-,n}$ converge uniformly on compact intervals (cf. Theorem 1 in Chapter II.8 of [Nikiforov and Uvarov \(1988\)](#)). Hence, we have also uniform convergence of f_n and F_n . Thus, boundedness of $\tilde{C}_n(\alpha)$ follows for all α if $f_{T|X}$ or $f_{C|X}$ are compactly supported because by definition $1 - F_n(y - x'\beta_\tau) > 0$. In order to have that $C_n(\alpha)$ is bounded, we note that f_n has likely zeros of degree 2 only. Therefore, the integrand in $C_n(\alpha)$ has singularities of degree $2\alpha/(1-\alpha) < 1$ when $\alpha < 1/3$. Hence, the singularities are integrable. Since the integrand is also multiplied by $f_{T|X}(y|x) = \varphi_0(y - x'\beta)$, we have the boundedness of $C_n(\alpha)$ for $\alpha \in (0, 1/3)$ if we assume a mild tail condition on φ_0 .

Remark 4.4. Let $\kappa, p \in \mathbb{N}$ and suppose that p_+ and p_- are 2κ -times differentiable. Proposition 7.1 in [Comte and Genon-Catalot \(2015\)](#) shows that $\delta_{+,n} = O(m_n^{-2\kappa})$ if $p_+(u|x) = p_+(u)$ does not depend on x , is 2κ times differentiable with $\int_0^\infty u^{m/2} p_+^{(m)}(u) e^{-u} du < \infty$ for all $m = 1, \dots, 2\kappa$ (and similarly for p_-), and if we were not enforcing a constraint on the weights. However, this constraint only ensures that the approximating polynomial induces a density function. Therefore, it seems reasonable to assume that also our formulation with Laguerre polynomials can achieve the same rates. Thus, if we let $m_n, \tilde{m}_n = O(n^\xi)$ for some $\xi > 0$, we have in view of Lemma 4.1 that $\delta_n(\alpha) \approx n^{-2\kappa\xi\alpha}$. In general, there are polynomials of degree $2^\kappa p$ which approximate p_+ and p_- in sup-norm on a compact interval with order $p^{-\kappa}$ (cf. Theorem 1 and Section 4 in [Butzer \(1953\)](#) and Theorem 9.1 in [Bustamante \(2017\)](#)). These particular approximating polynomials could be represented by weighted sums of any polynomial basis. Assuming that restricting to densities does not change the rate motivates similarly that $\delta_n(\alpha) = p^{-2\kappa\alpha} = n^{-2\kappa\xi\alpha}$ when $m_n = 2^\kappa p = O(n^{-\xi})$. Suppose, in addition, that $\beta_\tau \in K_n = K$, i.e., K_n is constant. For Theorem 4.2 to hold, we require that $\varphi_n \approx n^{\xi+\delta}$ for some small $\delta > 0$. When looking at the rate r_n , we see that $\xi > 0$ is subject to a trade-off: Choosing $\xi > 0$ small makes φ_n/n small, but, on the other hand, a large ξ increases the approximation quality of the sieve and hence $\delta_n(\alpha)$ is smaller. The best rate is obtained when both expressions in the maximum in the definition of r_n are equal, i.e., if

$$\frac{\varphi_n}{n} \approx \delta_n(\alpha) \Rightarrow \xi = \frac{1-\delta}{1+2\kappa\alpha} \Rightarrow r_n \approx n^{-\kappa\alpha \frac{1-\delta}{1+2\kappa\alpha}}.$$

In particular, we can obtain for example the following rates:

Parameter	Example 1	Example 2	Example 3
α	$\frac{1}{4}$	$\frac{15}{48}$	0.9
κ	7	7	3
δ	$\frac{1}{7}$	0.01	0.1
ξ	$\frac{4}{21}$	$\frac{198}{1075}$	$\frac{9}{64}$
r_n	$n^{-\frac{1}{3}}$	$n^{-\frac{693}{1720}}$	$n^{-\frac{243}{640}}$

The rate $n^{-1/3}$ in the first example will be too slow for later results. But the rates from Examples 2 and 3 will be fast enough. Examples 1 and 2 require a very large degree for general polynomials. But they allow for $\alpha < 1/3$ which is in almost all situations a good choice (cf. Remark 4.3). For Laguerre polynomials, the required degree depends only on ξ . In Example 3, we reduced the degree of the approximating polynomials but α lies close to one. In view of Remark 4.3, this is then reasonable if we can be sure that the approximating polynomials have no zeros. Note finally that for the case $\delta = 1/\kappa$,

$$-\frac{1}{2} < -\kappa\alpha \frac{1-\delta}{1+2\kappa\alpha} = -\alpha \frac{\kappa-1}{1+2\kappa\alpha} \xrightarrow{\kappa \rightarrow \infty} -\frac{1}{2}.$$

Hence, r_n is (unsurprisingly) always worse than the parametric rate, but the larger κ gets, the closer we can come to the $\sqrt{n}$ -rate.

Theorem 4.2 implies also that the estimate for the parameter β converges. Even though Remark 4.4 above shows that the convergence rate of the density itself cannot achieve a $\sqrt{n}$ -rate, we can show a $\sqrt{n}$ -rate for the quantile in Section 4.2 because when proving Lemma 4.10 the convergence rate of the density appears only in a second order Taylor remainder term, which *increases* the convergence rate. Such a phenomenon typically occurs, cf. Shen (1997) Theorem 1 and (4.2). For the moment we mention that $\hat{\beta}_n$ is consistent under the following additional assumption.

Assumption (A2). $\beta'X = 0$ almost surely implies $\beta = 0$, and the distribution function $F_{T|X}(\cdot|x)$ is bijective in a neighborhood of the true quantile $x'\beta_\tau$ for all $x \in \mathcal{X}$. Moreover, $F_{C|X}(\beta'_\tau X|X) < 1$ almost surely.

Assumption (A2) is natural as it is intuitively clear that the true quantile cannot be reconstructed from the censored observations if, e.g., all censoring happens before the quantile or if $\beta'_\tau x$ takes too few values. The bijectivity of $F_{T|X}$ is required to make sure that the quantile is well defined.

Corollary 4.5. *Let the conditions of Theorem 4.2 and (A2) hold. Then, $\hat{\beta}_n - \beta_\tau = o_P(1)$.*

4.2. Asymptotic normality

In this section, we will discuss the asymptotic normality of $\hat{\beta}_n$. Proofs of the results are provided in Section S.2 of the Supplementary Material (Kreiss and Van Keilegom, 2025). To this end, we require some additional notation. Let $g(y|x)$ be an arbitrary conditional density. If we suppose that the conditional distribution of T given X has a density g , the joint density of (Y, X, Δ) is given by

$$h_g(y, x, \delta) = f_X(x) \left[\delta g(y|x) (1 - F_{C|X}(y|x)) + (1 - \delta) f_{C|X}(y|x) \int_y^\infty g(t|x) dt \right].$$

In this notation, the true joint density of (Y, X, Δ) is given by $h_0 = h_{f_{T|X}}$, and for the estimator it holds that $\hat{h}_n = h_{\hat{f}_n}$ with $\hat{f}_n(y|x) = f_{\hat{\theta}_n, \hat{\theta}_n}(y - \hat{\beta}'_n x)$. Let now

$$\mathcal{D}_0 := \bigcup_{m, \tilde{m}=1}^{\infty} \mathcal{F}_{m, \tilde{m}} - f_{T|X} \quad \text{and} \quad \mathcal{D} := \text{span}(\mathcal{D}_0). \quad (4.3)$$

Then, $u \in \mathcal{D}_0$ should be understood as a direction of deviation from the truth $f_{T|X}$. We may define h_u analogously to h_g by replacing all g in the definition by u . Then, h_u is no longer a density function, and we understand h_u as a direction of deviation from $h_0 = h_{f_{T|X}}$. Note that we have linearity, that is, $h_{f_{T|X}+u} = h_0 + h_u$. We call $W(t; h_u) := h_0 + t h_u = (1-t)h_0 + t h_{f_{T|X}+u}$ the path from h_0 in direction h_u . Note that $W(t; h_u)$ is a joint density for all $t \in [0, 1]$ and $u \in \mathcal{D}_0$. With this notation, the directional derivative of $\ell(h; y, x, \delta) := \log h(y, x, \delta)$ at h_0 in direction h_u will be denoted as

$$\ell'_{f_{T|X}}[h_u; y, x, \delta] := \lim_{t \rightarrow 0} \frac{\ell(W(t; h_u); y, x, \delta) - \ell(h_0; y, x, \delta)}{t} = \frac{h_u}{h_0}. \quad (4.4)$$

Note that $\ell'_{f_{T|X}}[h_u; y, x, \delta]$ is linear in u . Therefore, we may extend $\ell_{f_{T|X}}$ to the linear span $\mathcal{D}$. For the extension, the interpretation as derivative is no longer correct, but we may understand the last equality above as definition. This allows us to equip the space $\mathcal{D}$ (which is linear) with a scalar product, which is often denoted as Fisher scalar product: Let $u, v \in \mathcal{D}$ and define

$$\begin{aligned} \langle u, v \rangle_F &:= \langle h_u, h_v \rangle_F := \mathbb{E} \left(\ell'_{f_{T|X}}[h_u; Y, X, \Delta] \cdot \ell'_{f_{T|X}}[h_v; Y, X, \Delta] \right) \\ &= \mathbb{E} \left(\frac{h_u(Y, X, \Delta) h_v(Y, X, \Delta)}{h_0(Y, X, \Delta)^2} \right). \end{aligned}$$

The corresponding norm is given by $\|u\|_F^2 := \|h_u\|_F^2 := \langle h_u, h_u \rangle_F$.

We follow [Patilea and Van Keilegom \(2020\)](#) to show how β_τ can be computed from h_0 . The conditional hazard rate of T is given by

$$\lambda_{T, h_0}(t|x) := \frac{h_0(t, x, 1)}{\int_t^\infty (h_0(a, x, 0) + h_0(a, x, 1)) da} = \frac{f_{T|X}(t|x)}{1 - F_{T|X}(t|x)} = -\frac{d}{dt} \log(1 - F_{T|X}(t|x)).$$

By means of the first equality, we may define a function $h \mapsto \lambda_{T, h}$ for any joint density h of (Y, X, Δ) whose marginal over X equals f_X . But the last two equalities hold only in the case $h = h_0$. Hence,

$$F_{T|X}(t|x) = 1 - \exp \left(- \int_{-\infty}^t \lambda_{T, h_0}(r|x) dr \right). \quad (4.5)$$

One can derive a similar representation of $F_{C|X}$ ([Patilea and Van Keilegom, 2020](#)). We obtain that the τ -th conditional quantile of T can be represented as, letting $\Lambda_{T, h_0}(q|x) := \int_{-\infty}^q \lambda_{T, h_0}(r|x) dr$,

$$\begin{aligned} \tau = F_{T|X}(\beta'_\tau x|x) &= 1 - e^{-\Lambda_{T, h_0}(x' \beta_\tau|x)} \Rightarrow x' \beta_\tau = \Lambda_{T, h_0}^{-1}(-\log(1 - \tau)|x) \\ &\Rightarrow \beta_\tau = Q(h_0) := \Sigma^{-1} \int_{\mathcal{X}} x \Lambda_{T, h_0}^{-1}(-\log(1 - \tau)|x) f_X(x) d\mu(x), \end{aligned} \quad (4.6)$$

where $\Sigma := \mathbb{E}(XX')$. Since $h \mapsto \lambda_{T, h}$ may be understood as a function of joint densities, we may also consider $h \mapsto \Lambda_{T, h}(q|x)$ and $h \mapsto Q(h)$. We call $Q(h)$ the quantile parameter of h . Let $\beta_t(u) := Q(W(t; h_u))$, for $u \in \mathcal{D}_0$, be the quantile parameter corresponding to $W(t; h_u)$, i.e., $\beta_0(u) = \beta_\tau$.

The directional derivative behaves well under the following assumption.

Assumption (AN1*). $\Sigma := \mathbb{E}(XX')$ is well-defined and invertible and there are $\eta_0, \alpha, C \in (0, \infty)$ such that for all $x \in \mathcal{X}$

$$f_{T|X}(x' \beta_\tau | x) \geq \eta_0, \quad F_{C|X}(x' \beta_\tau | x) \leq 1 - \alpha \text{ and } \|\Sigma^{-1}x\| \leq C.$$

It is intuitively plausible that (AN1*) is necessary for inference because it requires that the distribution function is not flat at the quantile. The condition on $F_{C|X}$ is a uniform extension of Assumption (A2) from Section 4.1 and has been discussed there. Invertibility of Σ is a classical assumption in regression, and the boundedness follows, e.g., for bounded $\mathcal{X}$.

Lemma 4.6. *If (AN1*) holds, the mapping $\mathcal{A} : \mathcal{D} \rightarrow \mathbb{R}^{1+p}, u \mapsto \lim_{t \rightarrow 0} t^{-1} (\beta_t(u) - \beta_0(u))$ is linear and bounded on the unit circle in the sense that for each $k \in \{1, \dots, 1+p\}$ there is $C > 0$ such that $|\mathcal{A}_k(u)| \leq C\|u\|_F$, where $\mathcal{A}_k$ is the k -th component of $\mathcal{A}$.*

Lemma 4.6 implies that we can extend $\mathcal{A}$ continuously and linearly to $\overline{\mathcal{D}}$, the closure of $\mathcal{D}$ with respect to $\|\cdot\|_F$. This extension can be written in a convenient way:

Corollary 4.7. *Under (AN1*), there is a function $u^* : \mathbb{R} \times \mathcal{X} \times \{0, 1\} \rightarrow \mathbb{R}^{1+p}$ such that each component belongs to $\overline{\mathcal{D}}$ and such that for all $u \in \overline{\mathcal{D}}$, it holds that $\mathcal{A}(u) = \langle u, u^* \rangle_F$, where the scalar product here denotes the vector of the scalar products of u with each component of u^* . u^* is sometimes called Riesz-Representer. In particular this means that for all $u \in \mathcal{D}$*

$$\lim_{t \rightarrow 0} \frac{\beta_t(u) - \beta_0(u)}{t} = \langle u, u^* \rangle_F. \quad (4.7)$$

Proof. Lemma 4.6 shows that each component of $\mathcal{A}$ is a linear, continuous mapping from the Hilbert space $\overline{\mathcal{D}}$ to $\mathbb{R}$. Therefore, the Riesz Representation Theorem guarantees the existence of u^* . $\square$

The function u^* will determine the variance of the estimator. We will provide an explicit formula for u^* in Section 4.3 in a special case. However, we emphasize already here that in this special case and in general u^* depends on the unknown densities $f_{T|X}$ and $f_{C|X}$. Therefore, from a practical point of view it is no restriction that we provide no general formula for u^* in the general case because for practical purposes the variance has to be estimated anyway. With this notation, we can formulate the main result of this section. Let h_{u^*} be the vector with components $h_{u_k^*}$, where u_k^* is the k -th component of u^* .

Theorem 4.8. *Under the Assumptions (A1), and (AN0)-(AN5) presented below, it holds that*

$$\sqrt{n}(\hat{\beta}_n - \beta_\tau) \xrightarrow{d} \mathcal{N}(0, V),$$

where $V = \mathbb{E} \left(\frac{h_{u^*}(Y, X, \Delta) h_{u^*}(Y, X, \Delta)'}{h_0(Y, X, \Delta)^2} \right)$, and u^* is defined in Corollary 4.7.

Proof. The theorem is a direct consequence of Lemmas 4.9 and 4.10 given below. More precisely, Lemma 4.10 gives for any $\gamma > 0$, since $h_{\gamma u^*} = \gamma h_{u^*}$,

$$\sqrt{n}(\hat{\beta}_n - \beta_\tau) = \sqrt{n} \langle \hat{h}_n - h_0, h_{u^*} \rangle + o_P(1) = \frac{1}{\gamma} \sqrt{n} \langle \hat{h}_n - h_0, h_{\gamma u^*} \rangle + o_P(1).$$

By Choosing the specific γ from Assumption (AN4), we may apply Lemma 4.9 to conclude

$$\sqrt{n}(\hat{\beta}_n - \beta_\tau) \xrightarrow{d} \frac{1}{\gamma} \mathcal{N}\left(0, \gamma^2 \mathbb{E}\left(\frac{h_{u^*}(Y, X, \Delta) h_{u^*}(Y, X, \Delta)'}{h_0(Y, X, \Delta)^2}\right)\right) = \mathcal{N}(0, V). \quad \square$$

Lemma 4.9. Under Assumptions (A1), (AN0)-(AN5), it holds for $\gamma > 0$ from Assumption (AN4) that

$$\sqrt{n} \langle \hat{h}_n - h_0, h_{\gamma u^*} \rangle_F \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}\left(\frac{h_{\gamma u^*}(Y, X, \Delta) h_{\gamma u^*}(Y, X, \Delta)'}{h_0(Y, X, \Delta)^2}\right)\right).$$

Lemma 4.10. Suppose that (A1), (AN1), (AN2) and (AN3, 4 and 5) hold. Then,

$$\sqrt{n}(\hat{\beta}_n - \beta_\tau) = \sqrt{n} \langle \hat{h}_n - h_0, h_{u^*} \rangle_F + o_P(1).$$

Note that Lemma 4.9 holds for an arbitrary choice of u^* which fulfills conditions (AN0)-(AN5); however, for Lemma 4.10 to hold we require the specific choice of u^* from Corollary 4.7. We state and discuss these assumptions in the following.

Assumption (AN0). The sets K_n are contained in a compact set K with $|K| \geq 1$ ($|K|$ denotes the diameter of K), and $m_n, \tilde{m}_n \leq cn^\xi$ for some $c > 0$ and $\xi \in (0, 1)$. It holds that $\mathbb{P}(\rho(\hat{h}_n, h_0) \geq r_n) \rightarrow 0$.

Rather than assuming $\mathbb{P}(\rho(\hat{h}_n, h_0) \geq r_n) \rightarrow 0$, we could of course also make the assumptions of Theorem 4.2.

Assumption (AN1). Additionally to the assumptions in (AN1*), the survival time T is lower bounded by T_0 , and the censoring time $C \in [T_0, C_0]$ for some $C_0 \in (0, \infty)$ (both almost surely). For almost all $x \in \mathcal{X}$, $t \mapsto f_{T|X}(t|x)$ is differentiable up to a countable set, and $0 < \eta_0 \leq f_{T|X}(y|x) \leq \eta_0^{-1}$ and $F_{T|X}(C_0|x) \leq 1 - \eta_0$ for all $y \in [T_0, C_0]$ and $x \in \mathcal{X}$. Furthermore, $\lambda_C(c|x) := f_{C|X}(c|x)/(1 - F_{C|X}(c|x))$ is uniformly bounded.

(AN1) strengthens Assumption (AN1*). Survival times T are naturally bounded from below by $T_0 = 0$. When censoring is linked to end of observation, it is also natural that the length of the observation period upper bounds the censoring time C . The requirement that C is bounded from below by T_0 is no restriction and can always be achieved. The positivity of $f_{T|X}$ on $[T_0, C_0]$ is also not too restrictive. The requirement $\sup_{x \in \mathcal{X}} F_{T|X}(C_0|x) < 1$ is of technical nature and simplifies the proofs.

Assumption (AN2). There is $\tilde{r}_n \rightarrow 0$ such that $\mathbb{P}\left(\sup_{y \in [T_0, C_0], x \in \mathcal{X}} |\hat{f}_n(y|x) - f_{T|X}(y|x)| \leq \tilde{r}_n\right) \rightarrow 1$ and $\mathbb{P}(\sup_{x \in \mathcal{X}} |\hat{F}_n(C_0|x) - F_{T|X}(C_0|x)| \leq \tilde{r}_n) \rightarrow 1$.

Recall that we have shown in Theorem 4.2 that we have convergence in the Hellinger distance. Therefore, a smoothness assumption on the true density implies also uniform convergence with a slower rate. Typically, we would expect that $\tilde{r}_n \approx r_n (\log n)^\gamma$, where $\gamma > 0$ depends on the level of smoothness, cf. Tsybakov (2009). In Remark 4.4, we gave a heuristic argument showing that we may expect (AN2) to hold.

Assumption (AN3). Let $\gamma_n := 1/\log n$ and $\varepsilon_n := \gamma_n/\sqrt{n}$. There is $\zeta > 0$ such that

1. $\sqrt{n} \tilde{r}_n / \gamma_n \rightarrow \infty$,

2. $\gamma_n \geq \tilde{r}_n$ and $\varepsilon_n \leq r_n$ both for n large enough,
3. $\tilde{r}_n^2 n^{\xi+\zeta} = O(1)$ for some $\zeta > 0$,
4. $\varepsilon_n^{-1} \tilde{r}_n r_n = O(1)$, and
5. $\sqrt{n} r_n^2 \max(m_n, \tilde{m}_n)^{\frac{3}{2}} \rightarrow 0$.

Recall that $r_n = n^{-\eta}$ with $\eta \in (0, 1/2)$ (cf. Remark 4.4), and suppose $\tilde{r}_n \approx r_n (\log n)^\gamma$. In this case 1., 2., and 3. are true. For 4., one requires $\eta \geq 1/4$, and 5. holds if $4\eta - 3\xi > 1$. Thus, (AN3) is for example fulfilled in the situations of Examples 2 and 3 in Remark 4.4. Note furthermore that Theorem 4.2 implies $\varepsilon_n \leq r_n$.

Consider for $\gamma > 0$ as in Assumption (AN4) below $f_n^* := (1 - \varepsilon_n) \hat{f}_n + \varepsilon_n (\gamma \cdot u^* + f_{T|X})$, and let also $h_n^* := h_{f_n^*} = (1 - \varepsilon_n) \hat{h}_n + \varepsilon_n (h_{\gamma u^*} + h_0)$. In general, the components of h_n^* are not members of $\mathcal{H}_n$ defined in Section 4.1. Let $\mathcal{J}$ denote the set of all possible joint densities, and let $P_n : \mathcal{J} \rightarrow \mathcal{H}_n$ be the projection with respect to the Hellinger distance, i.e., for any density $\alpha \in \mathcal{J}$ it holds that $\rho(\alpha, P_n(\alpha)) \leq \rho(\alpha, \tilde{h}_n)$ for all $\tilde{h}_n \in \mathcal{H}_n$. Hence, $P_n(h_{f_n^*}) \in \mathcal{H}_n$ (here we make abuse of notation and denote by $P_n(h_{f_n^*})$ the vector of the projections of each component), and we find $\bar{f}_n^* : \mathbb{R} \times \mathcal{X} \rightarrow \mathbb{R}^{1+p}$ such that each of its components is contained in $\mathcal{F}_{m_n, \tilde{m}_n}$ and such that $P_n(h_{f_n^*}) = h_{\bar{f}_n^*}$ (component-wise). For these projections, we require:

Assumption (AN4). There is $\gamma > 0$ such that $\gamma \cdot u^*(t|x) + f_{T|X}(t|x) \geq 0$, and its support w.r.t. t lies in $[T_0, \infty)$ for all $x \in \mathcal{X}$. Let $\bar{F}_n^*(y|x) = \int_{-\infty}^y \bar{f}_n^*(t|x) dt$ (component-wise). For each component $\bar{f}_{n,k}^*$ of $\bar{f}_n^*$ and each component $\bar{F}_{n,k}^*$ of $\bar{F}_n^*$ ($k = 1, \dots, 1+p$),

$$\mathbb{P} \left(\sup_{\substack{y \in [T_0, C_0], \\ x \in \mathcal{X}}} |\bar{f}_{n,k}^*(y|x) - f_{T|X}(y|x)| \leq \tilde{r}_n \right) \rightarrow 1 \quad \text{and} \\ \mathbb{P} \left(\sup_{x \in \mathcal{X}} |\bar{F}_{n,k}^*(C_0|x) - F_{T|X}(C_0|x)| \leq \tilde{r}_n \right) \rightarrow 1.$$

The above remains true when u^* in the definition of h_n^* is replaced by $-u^*$.

By definition, $f_n^* - \hat{f}_n \approx O(\varepsilon_n)$ and $\varepsilon_n = o(1/\sqrt{n})$. Intuitively speaking, (AN4) means therefore that perturbing $\hat{f}_n$ by a small amount and projecting back to $\mathcal{H}_n$ does not destroy the approximation properties that $\hat{f}_n$ has.

Assumption (AN5). It holds that (below, expectations are with respect to (Y_i, X_i, Δ_i) only)

$$\frac{1}{n} \sum_{i=1}^n \left(\ell'_{f_{T|X}}[h_n^* - P_n(h_n^*); Y_i, X_i, \Delta_i] - \mathbb{E}(\ell'_{f_{T|X}}[h_n^* - P_n(h_n^*); Y_i, X_i, \Delta_i]) \right) = O_P(\varepsilon_n^2), \quad (4.8)$$

$$\|P_n(h_n^*) - h_n^*\|_F^2 = O_P(\varepsilon_n^2), \quad \text{and} \quad (4.9)$$

$$\|P_n(h_n^*) - h_n^*\|_F \|\hat{h}_n - h\|_F = O_P(\varepsilon_n^2). \quad (4.10)$$

The above remains true when u^* in the definition of h_n^* is replaced by $-u^*$.

(AN5) requires that the perturbation h_n^* does not lie too far away from the sieve spaces. Since we keep u^* general at this point, we postpone a discussion of this assumption to Section 4.3 where we discuss a specific example.

4.3. Efficiency

We argue now that our estimator is efficient in the special case of no covariates, i.e., $X \equiv 1$. In this case, we may replace all conditional densities $f_{T|X}$ and $f_{C|X}$ by f_T and f_C , respectively. Moreover, in the results from Section 3, we may remove the dependence of x in all functions and let $g_X \equiv 0$. Note that in (4.4) we compute the score function of the linear path $W(t; h_u)$. Comparing (4.4) with Lemma 3.3 shows that the linear path $W(t; h_u)$ has the score $g_T = u/f_T$ and $g_C \equiv 0$. Using this in equation (S.3.13) (proof of Theorem 3.4) as well as the representation of Ψ_0 in (S.3.12) in the Supplementary Material (Kreiss and Van Keilegom, 2025), we obtain

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{\beta_t(u) - \beta_0(u)}{t} &= \int_{-\infty}^{\infty} \Psi_1(y) u(y) (1 - F_C(y)) dy + \int_{-\infty}^{\infty} \Psi_0(y) \int_y^{\infty} u(t) dt f_C(y) dy \\ &= \int_{-\infty}^{\infty} \left(\frac{\Psi_1(y) f_T(y) u(y)}{f_T(y)} (1 - F_C(y)) + \frac{\int_y^{\infty} \Psi_1(r) f_T(r) dr \cdot \int_y^{\infty} u(t) dt}{1 - F_T(y)} f_C(y) \right) dy \\ &= \mathbb{E} \left(\frac{h_{\Psi_1 f_T}(Y, \Delta) h_u(Y, \Delta)}{h_0(Y, \Delta)^2} \right) = \langle u, \Psi_1 f_T \rangle_F. \end{aligned}$$

Hence, (4.7) holds for

$$u^*(y) := \Psi_1(y) f_T(y) = -\frac{1 - \tau}{f_T(\beta_\tau)} (\mathbb{1}(y \leq \beta_\tau)(I(y) + 1) + \mathbb{1}(y > \beta_\tau)J(\beta_\tau)) f_T(y). \quad (4.11)$$

We need to argue that $u^* \in \overline{\mathcal{D}}$. Note firstly that $\int_{-\infty}^{\infty} u^*(y) dy = 0$. Therefore, by Assumption (AN1), we may choose $\gamma > 0$ so small such that $g(y) := \gamma u^*(y) + f_T(y) \geq 0$ is a probability density. Next, we note that $\mathcal{F}_{m, \tilde{m}}$ can be used to approximate densities that are continuously differentiable up to a discontinuity at the quantile β_τ (cf. Remarks 4.3 and 4.4). Unfortunately, g has a discontinuity at β_τ , which is not its quantile. In fact, it can be shown that $\int_{\beta_\tau}^{\infty} g(y) dy > 1 - \tau$ (for any $\gamma > 0$, but the integral converges to $1 - \tau$ for $\gamma \rightarrow 0$). Therefore, in general, we do not have $g - f_T \in \mathcal{D}_0$. However, when we choose $\gamma > 0$ so small such that $\int_{-\infty}^{\beta_\tau} g(y) > \tau/2$, there is a differentiable probability density φ such that $2g - \varphi \geq 0$ which has the property that $\int_{-\infty}^{\beta_\tau} (2g(y) - \varphi(y)) dy = \tau$. The function $2g - \varphi$ is hence a probability density which has its quantile at β_τ and has a discontinuity at β_τ . We have discussed that such densities can be well approximated with functions from $\mathcal{F}_{m, \tilde{m}}$, and, therefore, $(2g - \varphi) - f_T \in \overline{\mathcal{D}}$. Furthermore, since we allow different quantiles in $\mathcal{F}_{m, \tilde{m}}$, we can approximate the differentiable probability density φ well with functions from $\mathcal{F}_{m, \tilde{m}}$, and, hence, $\varphi - f_T \in \overline{\mathcal{D}}$. Hence, overall we have

$$u^* = \frac{1}{2\gamma} [\{(2g - \varphi) - f_T\} + \{\varphi - f_T\}] \in \overline{\mathcal{D}}.$$

Hence, we have found an explicit expression for u^* from Corollary 4.7 in the special case $X \equiv 1$. Note finally that the variance of $\hat{\beta}_n$ from Theorem 4.8 equals in this case, using the representation of Ψ_0 in (S.3.12) in the Supplementary Material (Kreiss and Van Keilegom, 2025),

$$v^2 = \mathbb{E} \left(\frac{h_{u^*}(Y, \Delta)^2}{h_0(Y, \Delta)^2} \right) = \mathbb{E} \left(\Delta \frac{u^*(Y)^2}{f_T(Y)^2} + (1 - \Delta) \frac{\left(\int_Y^{\infty} u^*(r) dr \right)^2}{(1 - F_T(Y))^2} \right) = \mathbb{E} \left(\Delta \Psi_1(Y)^2 + (1 - \Delta) \Psi_0(Y)^2 \right) = v_0^2,$$

where v_0^2 is the optimal variance defined in Corollary 3.6. We have thus proven the following theorem.

Theorem 4.11. *Suppose that $X \equiv 1$ and that Assumptions (A1), (AN0)-(AN5) hold. In this case, the estimator $\hat{\beta}_n$ is asymptotically efficient/optimal in the sense that its asymptotic variance equals the optimal variance from Corollary 3.6.*

We discuss next the Assumptions (AN4) and (AN5) for u^* as in (4.11). We begin with (AN4). Note that Remark 3.5 implies that Ψ_1 is bounded. Therefore, existence of $\gamma > 0$ such that $\gamma u^* + f_T = (\gamma \Psi_1 + 1)f_T \geq 0$ is clear. The support assumption follows from this representation as well. Note furthermore that, under Assumption (AN2), we have that $f_n^* - \hat{f}_n = \varepsilon_n(\gamma u^* - f_T - \hat{f}_n) = O_P(\varepsilon_n)$ (uniformly), where $\varepsilon_n = o(1/\sqrt{n})$. Since we may expect that the projection $\bar{f}_n^*$ is only closer to f_n^* than $\hat{f}_n^*$, we conclude that $|f_n^* - \bar{f}_n^*| = o_P(1/\sqrt{n})$ (uniformly), which is faster than the example of $\tilde{r}_n$, which we have discussed after Assumption (AN3). Assumption (AN5) extends the previous considerations from the question of approximability to the rate of convergence of the best approximation to f_n^* by functions from $\mathcal{F}_{m_n, \bar{m}_n}$. For (4.9), the previously motivated rate of ε_n is sufficient. However, for (4.8) and (4.10), we require a slightly better approximation. The issue here is, similar as before, that, in general, f_n^* has a discontinuity at β_τ , but its quantile is not at β_τ . When restricting $\hat{f}_n$ to be differentiable, we could argue for example that there is a sequence φ_n of differentiable approximations to $\gamma u^* + f_T$ such that

$$f_n^* - \left((1 - \varepsilon_n)\hat{f}_n + \varepsilon_n\varphi_n \right) = \varepsilon_n(\gamma u^* + f_T - \varphi_n)$$

converges to zero faster than ε_n , and it is sufficient to approximate $(1 - \varepsilon_n)\hat{f}_n + \varepsilon_n\varphi_n$. But this is a differentiable function, which can be well approximated under smoothness conditions.

Remark 4.12. For general covariates, we can construct u^* similarly. However, it will be difficult to show that this u^* is an element of $\overline{\mathcal{D}}$ because then one also has to argue that the linear structure in the covariates in $\mathcal{D}_0$ is rich enough to approximate the resulting f_n^* . We leave this question therefore open for further research.

5. Empirical analysis

In Section S.5.2 of the Supplementary Material (Kreiss and Van Keilegom, 2025), we apply our methodology to a data set of COVID-19 patients from Galicia (North-West Spain) that is also analyzed in López-Cheda et al. (2021) and was provided by the regional public health authority de Saúde Pública (General Directorate of Public Health). In the following we will discuss the implementation of the estimator and a simulation study.

5.1. Implementation

We discuss computation of $(\hat{\beta}_n, \hat{\theta}_n, \hat{\theta}_n^*, \hat{\lambda}_n)$ for given values of $m, \bar{m}, M$. To address the issue from Remark 2.5, we represent $\theta, \bar{\theta}$ and λ_a , for $a = 1, \dots, p$ in spherical coordinates, fix their radii to 1, and optimize over the angles. This way, we constrain the optimization to $\|\theta\| = \|\bar{\theta}\| = \|\lambda_a\| = 1$ for $a = 1, \dots, p$. By Lemma S.4.2 of the Supplementary Material (Kreiss and Van Keilegom, 2025), we have that constraining all angles to $[0, \pi]$ removes the ambiguities from Remark 2.5. We may thus use algorithms for box constrained optimization. Moreover, $L(\beta, \theta, \bar{\theta}, \lambda)$ is not convex (because the Laguerre polynomials can have zeros and therefore the likelihood may have singularities). We handle this in the optimization, similarly to Zhang and Davidian (2008), simply by trying several starting values. Specifically, for a $S \in \mathbb{N}$ we use the following algorithm:

1. Compute an estimator $\hat{\beta}_0$ with $m = \tilde{m} = M = 0$ (then the likelihood is convex).
2. Choose random starting points $\theta_1, \dots, \theta_S, \tilde{\theta}_1, \dots, \tilde{\theta}_S$ and $\lambda_1, \dots, \lambda_S$ (here we abuse notation and denote by $\lambda_i = (\lambda_{i1}, \dots, \lambda_{ip})$ a tuple of vectors).
3. Optimize $L(\beta, \theta, \tilde{\theta}, \lambda)$ starting from each of the starting values $(\hat{\beta}_0, \theta_i, \tilde{\theta}_i, \lambda_i)$. The final estimator is given as the maximizer of all these individual optimization tasks.

Step 1 of the algorithm could be replaced by computing a different estimator, or using random $\beta_1, \dots, \beta_S$, similarly as in step 2 with $\theta, \tilde{\theta}$, and λ . To determine $m, \tilde{m}$ and M , we apply Cross-Validation as in De Backer, El Ghouh and Van Keilegom (2020) and Wang and Wang (2009). To this end, the data-set is split in K disjoint folds $F_1, \dots, F_K$. For given $m, \tilde{m}$, and M , we compute estimates $\hat{\beta}_{-j}$ for $j = 1, \dots, K$, where $\hat{\beta}_{-j}$ was computed ignoring the data in F_j . The cross-validation criterion is then given by

$$CV(m, \tilde{m}, M) := \sum_{j=1}^K \left(\sum_{i \in F_j} \Delta_i \right)^{-1} \sum_{i \in F_j, \Delta_i=1} \rho_\tau(Y_i - X_i' \hat{\beta}_{-j}),$$

where $\rho_\tau(u) = u(\tau - \mathbb{1}(u \leq 0))$ (check-function). Note that m ($\tilde{m}$) influences only the right (left) tail of the distribution, and M only σ^2 . Therefore, the algorithm optimizes $m, \tilde{m}$ and M independently. We then start from the simplest model $m = \tilde{m} = M = 0$ and increase the dimension until there is no further decrease in the cross-validation criterion. The algorithm works as follows:

1. Set $m = \tilde{m} = M = 0$.
2. Compute $CV(m, \tilde{m}, M)$, $CV(m+1, \tilde{m}, M)$, $CV(m, \tilde{m}+1, M)$ and $CV(m, \tilde{m}, M+1)$.
3. If

$$CV(m, \tilde{m}, M) \leq \min(CV(m+1, \tilde{m}, M), CV(m, \tilde{m}+1, M), CV(m, \tilde{m}, M+1)),$$

go to step 6, otherwise go to step 4.

4. Increase either $m, \tilde{m}$ or M by one, depending on which increase yields the lowest cross-validation criterion. If, e.g.,

$$CV(m+1, \tilde{m}, M) < \min(CV(m, \tilde{m}+1, M), CV(m, \tilde{m}, M+1)),$$

set $m = m+1$.

5. Go to Step 2.
6. The current values of $m, \tilde{m}$ and M are the selected model dimensions.

Note that, in Step 2, one can potentially reuse results from previous runs.

In order to compute confidence intervals, we compute bootstrap estimates of the variance as follows:

1. Compute $\hat{\beta}_n$ including choices for $m_n, \tilde{m}_n$, and M .
2. Repeat for $b = 1, \dots, B$:
 - a) Draw a sample $(Y_i^*, X_i^*, \Delta_i^*)_{i=1}^n$ with replacement from $(Y_i, X_i, \Delta_i)_{i=1}^n$.
 - b) Compute $\hat{\beta}_b^*$ based on $(Y_i^*, X_i^*, \Delta_i^*)_{i=1}^n$ using $m_n, \tilde{m}_n$, and M from Step 1.
3. Compute $\hat{V}_n$ as the empirical covariance matrix of $[\sqrt{n}(\hat{\beta}_b^* - \hat{\beta}_n)]_{b=1}^B$.

We use the resulting matrix $\hat{V}_n$ to compute confidence intervals for β_τ . These procedures are implemented in the R-package *QuantRegLaguerre*¹.

¹ Available on the authors github page <https://github.com/akreiss/QuantRegLaguerre> or in the supplementary material

5.2. Simulation study

We investigate the finite sample performance of our estimator empirically in four different data generating processes (DGPs). Let $\text{Lap}(\tau)$ for $\tau \in (0, 1)$ denote the asymmetric Laplace distribution defined below (1.3) (it has its τ -th Quantile at 0).

DGP 1 $T_i = \beta_0 + \beta_1 X_{0,i} + (0.2 + 2(X_{0,i} - 0.5)^2)\varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, 1) - \Phi^{-1}(\tau)$, and $X_{0,i} \sim \mathcal{N}(0, 1)$

DGP 2 $T_i = \beta_0 + \beta_1 X_{0,i} + (0.2 + 2(X_{0,i} - 0.5)^2)\varepsilon_i$ with $\varepsilon_i \sim \text{Lap}(\tau)$, and $X_{0,i} \sim \mathcal{N}(0, 1)$

DGP 3 $T_i = \beta_\tau + \varepsilon_i$ with $\varepsilon_i \sim \text{Lap}(\tau)$

DGP 4 $T_i = \beta_0 + \beta_1 X_{0,i} + \varepsilon_i$, where the conditional distribution of ε_i given $X_{0,i}$ is a shifted t-distribution with degrees of freedom given by $4(1 + e^{1.5X_{0,i}})$, where the shift is such that the τ -th conditional quantile (given $X_{0,i}$) of ε_i lies at zero.

Censoring is independent in all DGPs and $C_i \sim \mathcal{U}([0, 7])$ (uniform distribution). In DGPs 1, 2, and 4, we let $X_i := (1, X_{0,i})'$ be the covariates and $\beta_\tau = (\beta_0, \beta_1)'$ be the true quantile parameter. In DGP 3, we let $X_i := 1$ and $\beta_\tau := 2$. We consider $\tau = 0.5$, $\tau = 0.25$, and $\tau = 0.7$ under $n = 200$ and $n = 500$ independent observations. In (almost) all cases, we consider 5,000 Monte Carlo replicates. DGP1 equals Example 2 from Wang and Wang (2009) and after slight changes Example 2 in De Backer, El Ghouch and Van Keilegom (2020). DGP2 differs from DGP1 only in the heavier tailed noise (asymmetric Laplace). DGP3 has no covariates. Finally, DGP4 has a more complicated noise structure, where the covariates do not only impact the variance of the noise but the entire distribution. We compare our estimator, where m , $\tilde{m}$, and M are chosen via cross-validation (Laguerre), to the methods by Portnoy (2003) (Portnoy), Peng and Huang (2008) (HuangPeng), Wang and Wang (2009) (WangWang), Bang and Tsiatis (2002) (BangTsiatis), and De Backer, El Ghouch and Van Keilegom (2020) (DBVKEG). For the methods (WangWang) and (DBVKEG), we implemented a cross-validation procedure similar as for our estimator to find the tuning parameters. The R-code that is provided for the methods (WangWang) and (DBVKEG) does not obviously show how to handle an intercept only model. Therefore, these methods have not been applied to DGP3.

The simulation results are shown in Tables 1-4 and Section S.5.1 of the Supplementary Material (Kreiss and Van Keilegom, 2025). Overall, the performance of the estimators is quite similar. In the simple DGP 1, cf. Table 1, our estimator performs best in terms of root mean squared error (RMSE) and standard deviation (SD) for $\tau = 0.5$. For $\tau = 0.7$, its performance is comparable to DBVKEG in terms of SD. For the low quantile $\tau = 0.25$, our new estimator performs best. The *Optimal* SD in Table 1 is according to Corollary 3.6. Note that we study efficiency under no assumptions on the model. DGP1 is a linear model with heteroskedasticity. It seems, therefore, that all studied estimators make use of this specific structure and, hence, they show a smaller standard deviation. In DGP 2, all standard deviations increase, but the differences between different methods seem minor in terms of SD. We see the same effect regarding the optimal SD. In DGP3 (cf. Table 2), again, all methods perform similarly. Theorem 4.11 states that our proposed estimator achieves the asymptotic efficiency bound. This is confirmed empirically in Table 2. However, the other estimator achieve it too. In view of Remark 3.7, we confirm here empirically that in linear homoskedastic models an estimator which does not use knowledge of the linear model is not necessarily disadvantaged over estimators that use this knowledge. The performance of the estimators in models that are not linear heteroskedastic models is investigated with DGP 4 (cf. Table 3). Again all estimators perform roughly equally well and they achieve the asymptotic efficiency bound. This is what we would expect because DGP4 has no typical structure that can be exploited.

The performance results for the inference procedure based on the bootstrap outlined in Section 5.1 is shown in Table 4: It shows the estimated standard errors and the coverage of the corresponding (marginal) confidence intervals based on 200 Bootstrap replicates. We see that the procedure works

Table 1. DGP1: Estimated Root Mean Squared Error (RMSE) and Standard Deviation (SD) from simulations using 5000 repetitions (entries with * are based on 1000 repetitions).

n	Method	$\tau = 0.5$				$\tau = 0.7$				$\tau = 0.25$			
		RMSE		SD		RMSE		SD		RMSE		SD	
		$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$
200	Portnoy	0.16	0.31	0.15	0.30	0.17	0.34	0.16	0.32	0.17	0.35	0.17	0.34
	HuangPeng	0.17	0.33	0.15	0.30	0.20	0.37	0.17	0.32	0.18	0.36	0.17	0.34
	WangWang	0.15	0.28	0.14	0.28	0.15	0.30	0.15	0.30	0.18	0.32	0.17	0.32
	BangTsiatis	0.21	0.39	0.17	0.34	0.20	0.39	0.18	0.37	0.29	0.47	0.20	0.37
	DBVKEG	0.15	0.30	0.14	0.27	0.13	0.31	0.13	0.30	0.34	0.39	0.25	0.35
	Laguerre	0.13	0.27	0.13	0.26	0.14	0.31	0.14	0.29	0.16	0.31	0.16	0.30
	Optimal			0.4	0.8			0.4	0.9			0.4	0.9
500	Portnoy	0.11	0.22	0.10	0.19	0.12	0.23	0.10	0.20	0.11	0.22	0.11	0.21
	HuangPeng	0.12	0.24	0.10	0.20	0.14	0.26	0.11	0.20	0.12	0.24	0.11	0.21
	WangWang	0.11	0.19	0.10	0.19	0.10	0.19	0.10	0.19	0.13	0.21	0.11	0.21
	BangTsiatis	0.17	0.30	0.11	0.23	0.14	0.28	0.12	0.24	0.23	0.37	0.13	0.24
	DBVKEG	0.09	0.18	0.09	0.17	0.09	0.19	0.09	0.18	0.14*	0.22*	0.12*	0.21*
	Laguerre	0.09	0.18	0.08	0.17	0.12	0.21	0.10	0.19	0.10	0.20	0.10	0.19
	Optimal			0.3	0.5			0.3	0.5			0.3	0.6

reasonably well. Note that the performance for $n = 500$ appears slightly worse. We believe that the reason is that, in the bootstrap procedure, we do not select $m, \tilde{m}$, and M in each bootstrap replication step, but we choose the same values as from the initial estimator. For small n this is apparently not problematic, but for large n , when larger values of $m, \tilde{m}$, and M are chosen, this yields an underestimation of the variance and, hence, too short confidence intervals.

Finally, Table 5 shows the runtime per method to fit one instance in DGP1 for $n = 200$ (averaged over 10 runs) on a computer with an *Intel(R) Core(TM) i5-9500T CPU, 2.20GHz* (without parallel computing). Note that for *WangWang*, *Laguerre*, and *DBVKEG* the runtime includes cross-validation. Therefore, their runtime is higher. Overall, for fast estimation, *Portnoy*, *HuangPeng*, and *BangTsiatis* dominate. However, no method takes prohibitively long if one is just interested in a single estimation.

Table 2. DGP3: Estimated Root Mean Squared Error (RMSE) and Standard Deviation (SD) from simulations using 5000 repetitions.

Method	$n = 200$						$n = 500$					
	$\tau = 0.5$		$\tau = 0.7$		$\tau = 0.25$		$\tau = 0.5$		$\tau = 0.7$		$\tau = 0.25$	
	RMSE	SD	RMSE	SD	RMSE	SD	RMSE	SD	RMSE	SD	RMSE	SD
Portnoy	0.16	0.16	0.18	0.18	0.19	0.19	0.10	0.10	0.11	0.11	0.12	0.12
HuangPeng	0.17	0.16	0.20	0.18	0.20	0.20	0.11	0.10	0.12	0.11	0.13	0.12
BangTsiatis	0.21	0.17	0.20	0.19	0.43	0.23	0.15	0.11	0.12	0.12	0.37	0.15
Laguerre	0.17	0.17	0.19	0.19	0.21	0.21	0.11	0.11	0.12	0.12	0.13	0.13
Optimal		0.15		0.17		0.18		0.10		0.11		0.11

Table 3. DGP4: Estimated Root Mean Squared Error (RMSE) and Standard Deviation (SD) from simulations using 5000 repetitions (the line with * is based on 1000 repetitions).

n	Method	$\tau = 0.5$				$\tau = 0.7$				$\tau = 0.25$			
		RMSE		SD		RMSE		SD		RMSE		SD	
		$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$	$\hat{\beta}_n^{(1)}$	$\hat{\beta}_n^{(2)}$
200	Portnoy	0.05	0.05	0.05	0.05	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06
	HuangPeng	0.06	0.06	0.05	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06
	WangWang	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.08	0.08	0.07	0.07
	BangTsiatis	0.05	0.06	0.05	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06
	DBVKEG	0.06	0.07	0.06	0.07	0.07	0.07	0.06	0.07	0.09	0.08	0.07	0.07
	Laguerre	0.09	0.08	0.09	0.08	0.10	0.09	0.10	0.09	0.08	0.08	0.08	0.08
	Optimal			0.05	0.05			0.03	0.03			0.08	0.08
500	Portnoy	0.03	0.03	0.03	0.03	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
	HuangPeng	0.04	0.03	0.03	0.03	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
	WangWang	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.05	0.06	0.04	0.05
	BangTsiatis	0.03	0.04	0.03	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
	DBVKEG*	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.07	0.05	0.04	0.05
	Laguerre	0.06	0.06	0.06	0.05	0.07	0.06	0.07	0.06	0.06	0.06	0.06	0.05
	Optimal			0.03	0.03			0.02	0.02			0.05	0.05

6. Conclusion

In this paper, we study quantile regression by extending the check-loss using Laguerre polynomials. We use these extended densities to build sieve spaces for the conditional survival density. The quantile is then estimated by the corresponding quantile of the sieve estimate. While we study the specific case on right-censored data, our method can quickly be extended to other forms of imperfect data. One reason why such a sieve approach might be appealing is that it allows to construct efficient estimators. We have, in addition, computed the optimal variance for quantile regression.

One interesting direction to continue the study of such problems is to understand in which scenarios the approximation through a linear quantile function is appropriate. Secondly, one can study further examples in which non-parametric density estimation can be used to estimate lower dimensional parameters from imperfect data. Next, in our proofs, we make in some places specific use of the fact that

Table 4. Bootstrap estimates for standard deviations (*B-SD*) and *Coverage* of the corresponding confidence intervals in DGP 1, 3, and 4.

n	τ	DGP1				DGP3				DGP4			
		B-SD		Coverage		B-SD		Coverage		B-SD		Coverage	
		β_1	β_2	β_1	β_2	β	β	β_1	β_2	β_1	β_2	β_1	β_2
200	0.5	0.14	0.27	95.70	94.52	0.19	95.60	0.08	0.08	90.00	94.08		
	0.7	0.14	0.28	93.76	93.30	0.20	95.12	0.09	0.10	88.60	94.34		
	0.25	0.16	0.30	95.56	94.82	0.23	95.48	0.07	0.08	90.36	95.92		
500	0.5	0.09	0.16	94.36	94.00	0.11	95.22	0.05	0.05	88.02	93.44		
	0.7	0.09	0.17	87.34	91.78	0.12	95.16	0.06	0.06	87.96	94.46		
	0.25	0.10	0.19	95.30	94.58	0.14	95.34	0.05	0.06	87.28	93.98		

Table 5. Time (seconds) required by each method to fit one instance of the data from DGP1 for $n = 200$.

	Portnoy	HuangPeng	WangWang	BangTsiatis	DBVKEG	Laguerre
Time in sec.	0.002	0.020	6.208	0.008	159.112	20.034

Laguerre polynomials are used. However, these places seem to be rather technical. Therefore, it would be interesting to better understand the role of Laguerre polynomials. Finally, we have seen that other estimators behave efficient too, and that all estimators outperform the efficiency bound in linear models. Hence, there seems to be adaptation to linear models. It is of interest to understand which estimator is efficient in which context and to which extend the existing estimators adapt to simpler situations.

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Supplementary Material

Proofs and additional empirical results (DOI: [10.3150/24-BEJ1829SUPPA](https://doi.org/10.3150/24-BEJ1829SUPPA); .pdf). In the Supplementary Material, we provide proofs for the results of Section 4.1, 4.2, and 3 in Sections S.1, S.2, S.3, respectively. Section S.4 collects useful results from the literature. Additional considerations for the empirical analysis and a real data application are provided in Section S.5.

Code (DOI: [10.3150/24-BEJ1829SUPPB](https://doi.org/10.3150/24-BEJ1829SUPPB); .zip). The file *QuantRegLaguerre.zip* contains the R-package that was used to compute the estimators based on Laguerre polynomials in Section 5.2.

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